DSML QUESTION BANK

MCA Semester III • Data Science & Machine Learning
Separate question sets for Modules 1–5
Built from your module notes, Series Test I and previous-year DSML papers

Complete question bank

This single PDF combines the question banks for all five DSML modules. Each module includes 4-mark questions, 8-mark questions, and numerical/diagram practice. The priority labels are based on your uploaded module notes, Series Test I, and previous-year DSML papers.

ANSWER TYPE	PREPARATION TARGET
4 marks	About 2 handwritten pages per answer.
8 marks	About 3–4 handwritten pages per answer.
Numericals	Show formula, substitution, intermediate calculations and final result.

MODULE 1 — DATA SCIENCE FUNDAMENTALS

Focus: Data Science process, statistics, datasets, descriptive/inferential statistics and visualization.

A. Short-answer questions — 4 marks

1. **[VH]** Define Data Science. Give one real-life example and explain how it supports decision-making.
2. **[VH]** What is a dataset? Explain its role in Data Science and Machine Learning.
3. **[VH]** Differentiate between descriptive and inferential statistics with examples.
4. **[VH]** Differentiate between univariate and multivariate analysis.
5. **[VH]** Explain the four types of analytics: descriptive, diagnostic, predictive and prescriptive.
6. **[H]** Differentiate classification and regression in supervised learning with examples.
7. **[H]** Define mean, median and mode. Explain their use in descriptive statistics.
8. **[H]** Define variance and standard deviation. What do they indicate about data?
9. **[H]** Explain the difference between AI, Machine Learning, Deep Learning, Big Data and Data Science.
10. **[H]** Explain the role of data preparation and data splitting in a Data Science workflow.
11. **[PYQ]** Explain ordinary least squares (OLS) estimation in regression and state the meaning of the fitted line. (Cross-check module placement.)

B. Long-answer questions — 8 marks

1. **[VH]** Describe the Data Science process with a neat labelled diagram. Explain each stage using a real-life example.
2. **[VH]** Explain visualization techniques for analyzing univariate and multivariate data, with suitable examples.
3. **[VH]** Explain common data visualization techniques and discuss which types of data each technique is suitable for.
4. **[H]** Explain the stages of a Data Science project from problem understanding and data exploration through modelling, deployment and monitoring.
5. **[H]** Explain descriptive and inferential statistics and compare their roles in data analysis.
6. **[H]** Explain univariate and multivariate analysis and discuss the plots used to explore distributions and relationships.
7. **[PYQ]** A regression equation is given: interpret its coefficients and intercept, and predict an output for supplied input values. (PYQ-linked; verify against class notes.)

C. Numerical and diagram practice

1. Given a set of marks, calculate mean, variance and standard deviation. Show all intermediate calculations.
2. Given a small dataset, identify the type of variable and select a suitable chart; justify your choice.
3. Given a regression equation and input values, calculate the prediction and interpret the coefficients if these are in your taught syllabus.

Module exam reminder

Use labelled diagrams where relevant, show all numerical working, and prepare PYQ-linked topics according to the official module mapping.

MODULE 2 — MACHINE LEARNING ALGORITHMS

Focus: learning types, KNN, probabilistic learning, Naive Bayes and Laplace smoothing.

A. Short-answer questions — 4 marks

1. **[VH]** Define Machine Learning and explain how a machine learns from data.
2. **[VH]** Explain data storage, abstraction, generalization and evaluation in machine learning.
3. **[VH]** Differentiate supervised, unsupervised, semi-supervised and reinforcement learning.
4. **[VH]** What is K-Nearest Neighbour (KNN)? Explain the basic steps of the algorithm.
5. **[VH]** What is the Naive Bayes classifier? State the assumptions on which it is based.
6. **[VH]** What is Laplace smoothing? Why is it used in Naive Bayes?
7. **[H]** Explain Euclidean distance and its role in KNN classification.
8. **[H]** Define prior probability, conditional probability and posterior probability.
9. **[H]** Differentiate classification and regression with one algorithm/example for each.
10. **[H]** State the advantages and limitations of KNN.
11. **[H]** Explain how rewards and penalties guide an agent in reinforcement learning.

B. Long-answer questions — 8 marks

1. **[VH]** Explain the KNN algorithm step by step and solve a classification problem using Euclidean distance for a specified value of k.
2. **[VH]** Explain the Naive Bayes classifier with Bayes' theorem and solve a classification example, showing prior, conditional and final class probabilities.
3. **[VH]** Using Naive Bayes with Laplace smoothing, predict the class of a new instance. Show the prior probabilities, smoothed conditional probabilities, joint scores and final class.
4. **[H]** Explain the four types of Machine Learning with their working principles, examples and suitable applications.
5. **[H]** Explain probabilistic learning and demonstrate how Bayes' theorem is used for classification.
6. **[H]** Explain the complete machine-learning learning cycle using data storage, abstraction, generalization and evaluation.
7. **[H]** Compare KNN and Naive Bayes in terms of working, assumptions, advantages and limitations.

C. Numerical and diagram practice

1. Use the weather dataset from the Series Test I paper to classify (Temperature = 24, Humidity = 65) with KNN and k = 3. Show distances, nearest neighbours and majority vote.
2. Use the film dataset from the Series Test I paper to predict Success = T or F for X = (Rest of world, Yes, Comedy) using Naive Bayes. Show class counts and conditional probabilities.
3. Use the PlayTennis table from the previous-year paper to predict a new instance with Naive Bayes and Laplace smoothing.
4. Given a labelled dataset and k, calculate Euclidean distances, rank them and predict the unknown class.

Module exam reminder

Use labelled diagrams where relevant, show all numerical working, and prepare PYQ-linked topics according to the official module mapping.

MODULE 3 — DECISION TREE LEARNING

Focus: entropy, information gain, Gini index, tree construction, pruning and classification rules.

A. Short-answer questions — 4 marks

1. **[VH]** Define entropy and explain what high and low entropy mean in a dataset.
2. **[VH]** Define Information Gain and write its formula.
3. **[VH]** What is the Gini Index? Write its formula and explain how it is used to select a split.
4. **[VH]** Define a Decision Tree. Explain root node, internal node, branch and leaf node.
5. **[VH]** Differentiate Information Gain and Gain Ratio.
6. **[VH]** Differentiate pre-pruning and post-pruning.
7. **[VH]** What is a classification rule? Explain antecedent and consequent with an IF–THEN example.
8. **[H]** Compare ID3, C4.5 and C5.0 based on splitting criterion and main capabilities.
9. **[H]** Explain why pruning is needed in Decision Tree learning.
10. **[PYQ]** What is feature selection and why is it necessary before building a Machine Learning model? (Verify module mapping.)

B. Long-answer questions — 8 marks

1. **[VH]** Calculate the entropy of a dataset and use Information Gain to select the best attribute for splitting. Show all calculations.
2. **[VH]** Explain the Gini Index in detail. Calculate the Gini impurity of a dataset and the weighted Gini for candidate splits, then choose the best split.
3. **[VH]** Explain how a Decision Tree is constructed from training data, using a suitable dataset and labelled diagram.
4. **[VH]** Explain in detail how a Decision Tree can be converted into a set of classification rules. Illustrate the conversion with a suitable tree and corresponding IF–THEN rules.
5. **[VH]** Explain pre-pruning and post-pruning. Compare their working, advantages and possible limitations.
6. **[H]** Explain ID3, C4.5 and C5.0 and compare their attribute-selection methods, pruning and handling of data.
7. **[H]** Explain the Divide-and-Conquer approach in Decision Tree learning and compare it with Separate-and-Conquer in classification-rule learning.
8. **[H]** Given a training table, calculate the initial entropy, weighted entropy and information gain for candidate attributes and select the root node.

C. Numerical and diagram practice

1. Calculate Gini(S) for a dataset with two or more classes using Gini(S) = 1 – Σ p■².
2. Calculate weighted Gini for two candidate splits and select the split with the lower weighted Gini.
3. Calculate entropy and Information Gain from class counts and subset counts; identify the best attribute.
4. Convert every root-to-leaf path of a supplied Decision Tree into one IF–THEN classification rule.

Module exam reminder

Use labelled diagrams where relevant, show all numerical working, and prepare PYQ-linked topics according to the official module mapping.

MODULE 4 — SUPPORT VECTOR MACHINE (SVM)

Core focus from the supplied module notes: hyperplane, margin, support vectors and classification. Additional previous-year topics are marked PYQ-linked because their exact module mapping should be verified.

A. Short-answer questions — 4 marks

1. **[VH]** Define Support Vector Machine (SVM) and explain its purpose in classification.
2. **[VH]** What is a hyperplane? Write the equation of a linear decision boundary and explain its terms.
3. **[VH]** Define support vectors and explain why they are important.
4. **[VH]** What is the margin in SVM? How does margin size affect generalization?
5. **[H]** Differentiate a decision boundary and a margin.
6. **[H]** Explain the role of weights and bias in the SVM hyperplane equation.
7. **[PYQ]** Explain the concept of feature selection and why it is needed before model building. (Cross-check module placement.)
8. **[PYQ]** Explain single-layer feedforward, multilayer feedforward and recurrent neural-network topologies with one application each. (Cross-check module placement.)
9. **[PYQ]** Explain the purpose of backpropagation and the role of learning rate. (Cross-check module placement.)

B. Long-answer questions — 8 marks

1. **[VH]** Explain the working of SVM with a neat diagram showing the hyperplane, margin boundaries, two classes and support vectors.
2. **[VH]** For a two-dimensional dataset, explain how the maximum-margin hyperplane separates two classes and why the support vectors determine the decision boundary.
3. **[VH]** Explain the concept of a support vector. Why are only support vectors crucial in defining the decision boundary while other data points are not?
4. **[H]** Explain linear separability, decision boundary, maximum margin and support vectors with a suitable example.
5. **[PYQ]** Explain different neural-network topologies—single-layer feedforward, multilayer feedforward and recurrent—and give one application of each. (Cross-check module mapping.)
6. **[PYQ]** Explain backpropagation for training neural networks, including error calculation, backward propagation, learning rate and weight updates. (Cross-check module mapping.)
7. **[PYQ]** Compare bagging, boosting and Random Forests in working principle, overfitting and suitability. (Cross-check module mapping.)
8. **[PYQ]** Given a linear regression equation, interpret the coefficients and predict a value for supplied inputs. (Cross-check module mapping.)

C. Numerical and diagram practice

1. Given a hyperplane equation w■x■ + w■x■ + b = 0 and a point, substitute the point and determine which side of the boundary it lies on.
2. Draw a two-class SVM diagram with the maximum-margin hyperplane, parallel margins and support vectors.
3. Practise any numerical or derivation explicitly shown in your SVM module notes.

Module exam reminder

Use labelled diagrams where relevant, show all numerical working, and prepare PYQ-linked topics according to the official module mapping.

MODULE 5 — CLUSTERING

Focus: clustering concepts, K-means, Euclidean distance, centroid updates, elbow method and hierarchical clustering.

A. Short-answer questions — 4 marks

1. **[VH]** Define clustering and explain how it differs from classification.
2. **[VH]** Explain the steps of the K-means clustering algorithm.
3. **[VH]** Write the Euclidean distance formula and explain its role in clustering.
4. **[H]** What is a centroid? How is a new centroid calculated in K-means?
5. **[H]** Explain the elbow method and its purpose in selecting the number of clusters.
6. **[H]** Differentiate agglomerative and divisive hierarchical clustering.
7. **[H]** What is a dendrogram? How is it used to interpret hierarchical clustering?
8. **[PYQ]** Explain why K-means may fail for non-spherical clusters and suggest ways to improve clustering. (Follow methods covered in class notes.)
9. **[H]** State the advantages and limitations of K-means.

B. Long-answer questions — 8 marks

1. **[VH]** Explain the K-means clustering algorithm with a worked example. Show initial centroids, distance calculations, cluster assignments, centroid updates and convergence.
2. **[VH]** Explain how the choice of distance measure and number of clusters (k) affects the quality of K-means clustering.
3. **[H]** Explain the elbow method with a suitable illustration and describe how it helps select k.
4. **[H]** Explain hierarchical clustering with a dendrogram. Compare agglomerative and divisive approaches.
5. **[H]** Compare K-means and hierarchical clustering in terms of working, output, advantages and limitations.
6. **[PYQ]** Explain why K-means may fail on non-spherical clusters and discuss possible improvements, limited to techniques covered in your course.
7. **[H]** Explain clustering applications and the role of distance measures in grouping similar observations.

C. Numerical and diagram practice

1. Perform one complete K-means iteration: calculate distances from each point to every centroid and assign points to the nearest centroid.
2. Calculate updated centroids using the mean of the coordinates of points assigned to each cluster.
3. Repeat assignment and centroid update until the cluster assignments no longer change.
4. Given a K-versus-WCSS table or plot, identify the likely elbow and justify the selected number of clusters.

Module exam reminder

Use labelled diagrams where relevant, show all numerical working, and prepare PYQ-linked topics according to the official module mapping.

FINAL CHECKLIST — BEFORE THE EXAM

■ Can I write a full 4-mark answer in roughly 2 handwritten pages without padding?

■ Can I write a full 8-mark answer in roughly 3–4 handwritten pages with a diagram or worked example where relevant?

■ Have I solved at least one KNN numerical, one Naive Bayes numerical with Laplace smoothing, one entropy/Information Gain calculation, one Gini calculation and one K-means numerical without looking at the solution?

■ Can I draw the Data Science process, a Decision Tree, an SVM margin diagram and a clustering/dendrogram illustration?

■ Have I revised definitions, comparisons, formulas, advantages, limitations and applications for every module?

■ Have I cross-checked PYQ-linked topics such as regression, neural networks, feature selection, confusion-matrix metrics and ensemble learning against the official module mapping?

Priority is based on the supplied DSML module notes, Series Test I and DSML question papers on pages 1–7 of S3 previous year question papers.pdf. It is a study guide, not a guarantee of exact exam questions. Follow the official syllabus and lecturer's module mapping where they differ.